

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z3,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple **Z3 ZL**: $\frac{Z_3Z_Lg_m}{Z_3g_m+Z_Lg_m}$

$$H(z)=\frac{Z_3Z_Lg_m}{Z_3g_m+Z_Lg_m}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s)=\left(\infty,\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \infty,\ R_L+\frac{1}{C_Ls}\right)$

$$H(s)=\frac{C_LR_3R_Ls+R_3}{C_3C_LR_3R_Ls^2+s(C_3R_3+C_LR_3+C_LR_L)+1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L}}{C_3R_3+C_LR_3+C_LR_L}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L}}$
bandwidth: $\frac{C_3R_3+C_LR_3+C_LR_L}{C_3C_LR_3R_L}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_LR_3R_L}{C_3R_3+C_LR_3+C_LR_L}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s)=\left(\infty,\ \infty,\ R_3+\frac{1}{C_3s},\ \infty,\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s)=\frac{C_3R_3R_Ls+R_L}{C_3C_LR_3R_Ls^2+s(C_3R_3+C_3R_L+C_LR_L)+1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L}}{C_3R_3+C_3R_L+C_LR_L}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L}}$
bandwidth: $\frac{C_3R_3+C_3R_L+C_LR_L}{C_3C_LR_3R_L}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_3R_3R_L}{C_3R_3+C_3R_L+C_LR_L}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, R_3, \infty, \infty, R_L)$

$$H(s) = \frac{R_3 R_L}{R_3 + R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3}{C_L R_3 s + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_3 R_L}{C_L R_3 R_L s + R_3 + R_L}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, \infty, R_3, \infty, \infty, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_3 R_L s + R_3}{s(C_L R_3 + C_L R_L) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L\right)$

$$H(s) = \frac{R_L}{C_3 R_L s + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s}\right)$

$$H(s) = \frac{1}{s(C_3 + C_L)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L}{s(C_3 R_L + C_L R_L) + 1}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty, \infty, \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_L s + 1}{C_3 C_L R_L s^2 + s(C_3 + C_L)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, R_L\right)$

$$H(s) = \frac{R_3 R_L}{C_3 R_3 R_L s + R_3 + R_L}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_3}{s(C_3 R_3 + C_L R_3) + 1}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty, \infty, \frac{R_3}{C_3 R_3 s + 1}, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_3 R_L}{R_3 + R_L + s(C_3 R_3 R_L + C_L R_3 R_L)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L \right)$

$$H(s) = \frac{C_3 R_3 R_L s + R_L}{s(C_3 R_3 + C_3 R_L) + 1}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_3 R_3 s + 1}{C_3 C_L R_3 s^2 + s(C_3 + C_L)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\infty, \infty, R_3 + \frac{1}{C_3 s}, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_3 C_L R_3 R_L s^2 + s(C_3 R_3 + C_L R_L) + 1}{s^2(C_3 C_L R_3 + C_3 C_L R_L) + s(C_3 + C_L)}$$

10 X-INVALID-WZ

11 X-PolynomialError